Open Design Project: Client Outline and Pitch
Team: Spotted… then Stopped
Clients: Cornell CALS Extension / E\&J Gallo Winery / National Grape

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Problem Statement:
New York vineyard operators aim to maximize yield during mechanical harvesting, but Spotted Lanternflies (SLF) frequently enter harvest bins and contaminate products. A Cornell–Penn State study found that 60% of lanternflies on vines end up in harvest bins, and current standards require rejection if even 1–2 adult insects are found per 1 kg sample. Significantly disturbing the vine to remove SLF's proves a problem when harvesting. Ripe grapes detach very easily, potentially resulting in high losses if they are directly exposed to excessive forces.

Impact:
Reducing contamination during the harvesting process leads to less wasted product and maximizes yield.

Proposed Direction:
Our main idea is to have a machine attached to the front of the harvester that would get rid of the flies right before the grapes are collected.

Concept: Front-Mounted Rotating Brush System (Figure 1 on second page)
An attachment to the front of the vehicle that has rotating brushes aimed at only the vine. The brushes would be located on a rotating rod (almost like a car wash brush) that scrapes the flies off the vine, placed at the right height so as to not disturb the grapes too much.

How it would be used:
- Mount system to harvester front
- Adjust height to match vine height
- Brushes rotate at speed matched to forward motion
- Flies are dislodged downward before grape collection

Why it's better than the status quo:
- Removes insects before contamination
- Avoids post-harvest filtering grapes and flies
- Compatible with current harvesting equipment

End-of-semester proof-of-concept:
Two cylindrical brushes attached to an adjustable mounting frame, spinning in tandem.

Key Risks / Unknowns:
- Damage to vines — Reduce yield if the vine is scraped: Test bristles on vine and assess damage.
- Knocking off grapes prematurely — Accidental removal of grapes: Test to ensure right height of bristles to avoid contact with grapes.
- Effectiveness of brushes — Might still leave some contamination: Test prototype on model of the vine with attachments of similar grip strength to SLF.

Questions for the client
1. Can we get a blueprint/design for the harvesters used?
    Affects how the brush would mount the harvester, and design considerations that come with that.
2. What is the average height/length of the vine?
    Affects the dimensions of the brush and the amount of materials needed.
3. What is the durability of the vine?
    Affects the amount of force we can put on the vine and the material used for the bristles.

References:

https://wineamerica.org/wp-content/uploads/2025/05/2025-WineAmerica-Methodology-4-28-25-Final.pdf

https://www.psu.edu/news/research/story/grapevines-may-only-need-help-survive-heavy-spotted-lanternfly-infestations

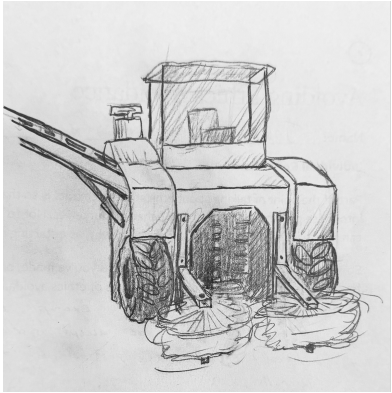

Figure 1: System mounts on the front of the harvester and brushes vines as it travels through the vineyard